

Car Rental Management System: Definitions and References

1. Key Definitions & Terminology

System Overview & Modules

- **Car Rental Management System:** A centralized web application acting as a digital bridge between car rental business operators and customers. It facilitates vehicle browsing, online booking, payment processing, fleet management, booking tracking, and business reporting.
- **Customer Module:** The module enabling customers to register an account, filter and search for vehicles (by brand, model, price range), book rentals, and handle billing automatically.
- **Admin Module:** The administrative interface providing operators/owners with tools to manage fleets, monitor customer accounts, track booking statuses, and generate usage and profit reports.

User Roles & Actors

Actor	Description
Guest	An unregistered person who has not yet logged into the system through the client website.
Registered Customer	A registered user who can manage account profiles, reserve vehicles, provide ratings/feedback, and process payments securely.
Administrator / Operator	Authorized personnel responsible for maintaining vehicle fleets, approving rental requests, and monitoring system health.
Payment Gateway	An external secondary actor/service responsible for securely processing

Actor	Description
	financial transactions and verifying payment completion.

Architecture & System Design Concepts

- **3-Layered Architecture:** An architectural pattern dividing the system into:
 - **Front End:** Handles the user interface and replaces paper-based workflows with digital interfaces.
 - **Back End:** Encapsulates business logic, data validation algorithms, and API integrations.
 - **Database:** Normalized relational database schema (MySQL) for structured data storage and referential integrity.
- **DFD Level 0 (Context Diagram):** The highest-level process diagram presenting the primary context of the system, depicting incoming and outgoing data flows between external entities (Admin, Customer) and the central system process.
- **DFD Level 1:** Broadens the context diagram into sub-processes (e.g., Manage Customer Information, Manage Car Information, Monitor Car Rentals, Manage Transactions).
- **DFD Level 2:** The most detailed level of data flow abstraction, incorporating specific databases (Customer, Car, Rental, Transaction) and low-level data inputs and outputs.
- **Entity Relationship Diagram (ERD):** Data modeling structure specifying entities (Users, Bookings, Cars, Payments, Categories) and foreign-key relationships to preserve referential integrity.
- **Domain Model / Class Model:** Object-oriented structural models mapping key system domain classes (User, Customer, Admin, Car/Vehicle, Reservation/Booking, Payment), their attributes, methods, and multiplicities.

Testing Definitions

- **Unit Testing:** Testing individual components/modules (such as the rental payment calculator) to verify correctness in isolation.
- **Integration Testing:** Testing verification of cross-module communication between the frontend, backend, and database tiers.
- **System Testing:** End-to-end evaluation ensuring the complete integrated application adheres to all functional and non-functional requirements specified in the SRS.

- **User Acceptance Testing (UAT):** The final verification phase conducted by stakeholders and end users to ensure the software functions reliably in a real-world car rental operational environment.

2. References

Formal Standards

- **IEEE Standard 830-1998:** Recommended Practice for Software Requirements Specifications (SRS).

Industry Benchmarks & Platforms

- **Uber & Careem:** Industry benchmarks used for real-time logistics, low-latency rental availability, fair/transparent pricing, and modern end-user booking experience.
- **Pak car rentals:** Existing paper/manual system reference.
- **New Pakistan motors:** Existing automotive rental market benchmark.
- **PakWheels:** Automotive marketplace and vehicle rental reference platform.
- **Rent A Car Services 24/7:** Online vehicle rental service reference (<https://www.rentacarservices247.com/>).